The Burger Battle: Inventory and Pricing under Competition

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Summary

Background & Formulation: BunBop and PattyPop compete selling identical burgers near a university campus. Each morning, both shops independently decide whether to open (paying an upfront fee of 4 coins), how many burgers to order from two suppliers (up to 4 from LittleFlame at 2 coins, then up to 8 from MegaGrill at 4 coins), and set an integer price between 1 and 8 coins. Demand varies randomly across quiet, normal, and festival days (6, 12, and 18 customers with maximum willingness-to-pay of 4, 6, and 8 coins, each occurring with probability 1/3). Unsold inventory expires with zero salvage value, and borrowing is prohibited. We formulate this environment as a stochastic inventory game to answer three central operational questions: finding the optimal single-day plan, designing an adaptive multi-period strategy across 30 days, and evaluating whether university students unambiguously benefit from duopoly rivalry.

Single-Day Game & The +4.00 Ceiling: With 30 starting coins, there are exactly 73 feasible operational plans. We completely enumerate all 73 x 73 pairings. Price ties are solved analytically via a closed-form binomial customer-overflow derivation, eliminating Monte Carlo error. We prove that the theoretical upper bound on expected daily profit is +4.00 coins, bounded by the profit of an unconstrained monopolist. The optimal plan--"open, buy 4 burgers from LittleFlame, and price at 6 coins" (buy 4 @ 6)--attains this +4.00 ceiling against every rival stocking up to 8 burgers. The plan breaks even on normal/festival days (+12 profit) while conceding quiet days (-12). All 39 pure-strategy Nash equilibria pay both shops exactly +4.00, and buy 4 @ 6 represents the unique, robust maximin strategy that guarantees +4.00 against all feasible initial rival profiles.

Thirty-Day Dynamics & Dual-Threshold Adaptation: Over 30 trading days, cash carries over and customer ties are simulated via 4,000 independent Monte Carlo paths. We reveal a critical structural hazard: the fixed one-day optimum suffers a 23.6% ruin probability under the Gambler's Ruin phenomenon (three consecutive quiet days wipe out initial liquidity). A passive stop-loss (shutdown below 12 coins) fails because permanent closure prevents cash recovery. To overcome this, we design the **Dual-Threshold Safe-Adaptive Strategy (E+)**: play 4 @ 6 when liquid, but dynamically switch to the safe-harbor pricing of 4 @ 4 whenever cash falls to <= 18 coins (where downside quiet-day losses are zero), restoring 4 @ 6 once cash refills; it simultaneously drops to 4 @ 4 if the rival stockpiles > 8 burgers. E+ **completely eliminates ruin (0.0% death rate)** across all regular rivals while boosting mean final cash to 127-143 coins. Crucially, price ranking dominates price level, and volume does not equal profit.

Student Welfare & Policy Paradox: Students do not always gain from market competition. While all 39 Nash equilibria yield identical profits (+4.00/day) to the shops, consumer surplus across these equilibria varies threefold, from 5.33 to 16.00 coins/day. In the symmetric high-price equilibrium (4@6, 4@6), student surplus is 5.33--substantially worse than under a single benevolent monopolist charging 4 coins (8.00). Market competition fails students through four distinct economic mechanisms: equilibrium miscoordination, quiet-day market abandonment, duopoly bankruptcy turning into surviving monopoly, and destructive price wars that collapse commercial viability.

Keywords: Stochastic Inventory Game; Minimax Optimality; Gambler's Ruin; Dual-Threshold Strategy; Consumer Surplus; Nash Equilibrium.

1 One-Day Competition

1.1 Setup and assumptions

BunBop and PattyPop each start the day with 30 coins and no stock [1]. Each shop chooses one plan before customers arrive: stay closed, or open by paying 4 coins, buy q burgers and set a price p between 1 and 8 coins. Both shops start from the same 30 coins, so maximising the final cash balance is the same as maximising the profit of the day; staying closed earns nothing.

Burgers come from LittleFlame at 2 coins each (at most 4 per day) and then from MegaGrill at 4 coins each (at most 8 more). Unsold burgers are worthless at closing time and borrowing is not allowed, so the opening fee plus the purchase cost cannot exceed the cash on hand: at most 8 burgers on the first day.

The day type is unknown when the plan is chosen. Quiet days bring 6 customers who pay at most 4 coins, normal days 12 customers at most 6 coins and festival days 18 customers at most 8 coins, each with probability 1/3. Customers arrive one at a time and go to the cheaper shop that still has burgers within their budget; at equal prices each customer picks a shop at random, and a customer who finds their first choice sold out may try the other shop. Our model reproduces the worked example in the problem statement: with 4 burgers at 4 coins the six quiet-day customers split evenly in expectation, so the expected profit is 4 x 3 - 4 - 8 = 0.

For this single day we recommend opening with 4 burgers from LittleFlame at a price of 6 coins, which earns an expected +4.00 coins and, as Section 1.4 shows, cannot be beaten.

Notations: Let D and w denote customer demand and willingness-to-pay. For shop i, $q_i \in \{0, \dots, 8\}$ denotes stock, $p_i \in \{1, \dots, 8\}$ price, s_i units sold, and $C(q) = 2\min(q, 4) + 4\max(0, q-4)$ the purchase cost.

1.2 How we searched

Only 73 plans are available on the first day, so we enumerated all of them: closing, plus every stock level q from 0 to 8 and every price p from 1 to 8 that the cash constraint allows. Each pair of plans has an exact expected profit, the average over the three day types, so this section needs no simulation.

The one random ingredient is the tie-break at equal prices. With D customers, let x_i ~ Binomial(D, 1/2) be the number who try shop i first; shop i then sells

$$(1)\ sold_i = min(x_i, q_i) + min(max(0, x_j - q_j), q_i - min(x_i, q_i)),$$

where the second term counts the customers turned away from the other shop. The formula matters: buy 8 @ 4 sells 3.125 burgers on average on a quiet day, which is what produces its -15.50. The resulting 73 x 73 payoff matrix gave best responses, pure Nash equilibria [3] and the maximin plan; Appendix A gives the derivation of (1).

1.3 Comparing candidate plans

Table 1 gives the expected profit of five plans against a benchmark rival that buys 4 burgers and charges 4 coins, the cautious strategy suggested in the problem statement.

Table 1: expected profit per day type against the benchmark rival.

Plan	Quiet	Normal	Festival	Expected
Buy 4 @ 4	0.00	+4.00	+4.00	+2.67
Buy 4 @ 6	-12.00	+12.00	+12.00	+4.00
Buy 8 @ 6	-28.00	+20.00	+20.00	+4.00
Buy 6 @ 4	-7.50	+4.00	+4.00	+0.17
Buy 8 @ 4	-15.50	+4.00	+4.00	-2.50

Two effects explain the ranking. First, overstocking is costly: four extra MegaGrill burgers cost 16 coins and produce only 2.71 extra sales in expectation, worth about 10.83 coins, so buy 8 @ 4 loses about 5.17 coins against buy 4 @ 4 and turns +2.67 into -2.50. Second, price 4 competes head-on: the quiet-day customers are shared with the rival, so buy 4 @ 4 earns +2.67 and the same price with more stock earns almost nothing (+0.17). Price 6 gives up the quiet day, a predictable -12, but earns +12 on each of the other two days, and at that price the stock level stops mattering: buy 4 @ 6 and buy 8 @ 6 both average +4.00.

Figure 1 repeats the experiment for every affordable plan. The +4.00 ceiling appears only in the p = 6 column, where it is reached at q = 4 and stays flat, so the cheapest plan on the plateau is the one to take. Every other price falls short: the p = 4 column becomes negative above six burgers, and prices 7 and 8 are negative at every stock level.

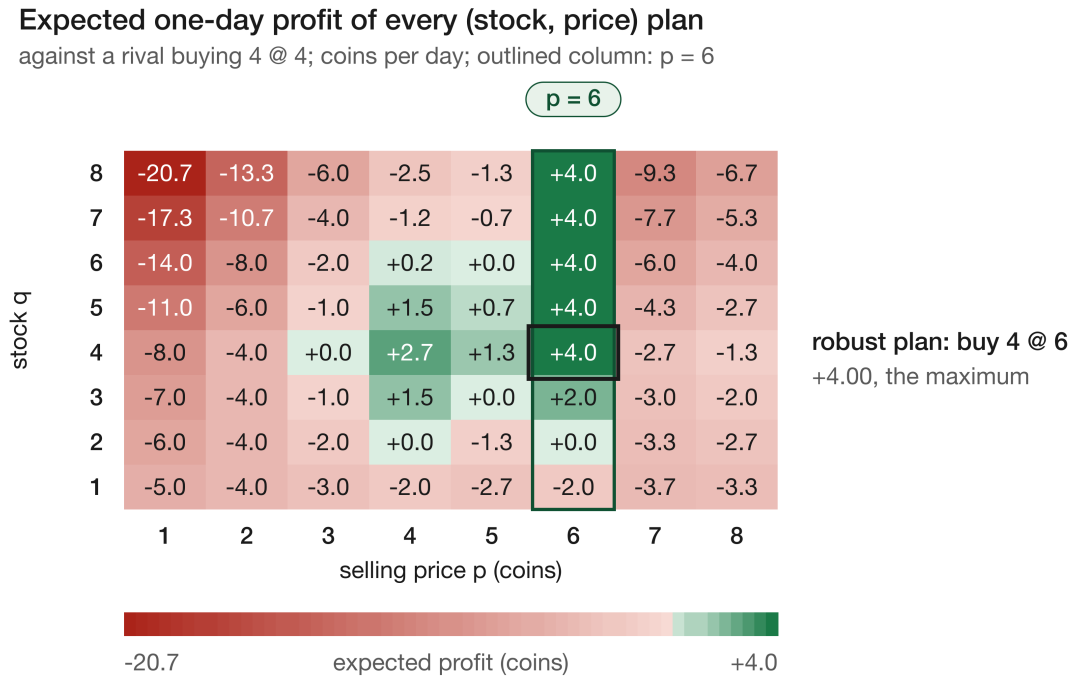


Figure 1: expected profit of every affordable (stock, price) plan against the benchmark rival. The p = 6 column is the only one reaching the +4.00 ceiling, and buy 4 @ 6 is the cheapest plan on that plateau.

1.4 The value of the one-day game is 4 coins

We can fix the value of the game without assuming anything about the rival. A shop can never sell more than min(D, q) burgers, its stock or the day's demand, whichever is smaller, and that is what it sells when the rival stays closed [2]. A rival can only take customers away, so a plan earns at most what the same plan earns as a monopolist, and the best monopoly profit is +4.00 over the affordable grid, and also over the unrestricted grid q <= 12, where it is attained at p = 6 for every q >= 4. The bound is attained: buying 4 burgers at 6 coins earns +4.00 against every feasible rival plan (Section 1.6). The maximin value of the one-day game is therefore exactly 4 coins, and no plan can earn more than that against any rival.

1.5 Why exactly 4: a knife-edge in price and stock

At 6 coins a burger sells only on normal and festival days, which occur with probability 2/3, so an extra MegaGrill burger costing 4 coins has expected marginal revenue 6 x (2/3) = 4 and adds nothing, while the first four burgers, at 2 coins each from LittleFlame, are worth stocking. Competition can break that flatness: against a rival buying 6 @ 6, buy 8 @ 6 earns only +1.16, because an equal-price rival shares the tie-broken demand.

Price 6 sits exactly at the normal-day budget, so the optimum is a knife-edge. At 7 coins the normal day is lost as well and the monopoly profit falls to -2.67, while at 4 coins the shop gives up 2 coins of margin on the two busy day types. Table 2 shows why that matters once a rival is present.

Table 2: the monopoly benchmark cannot separate the two sweet spots; competition can.

Plan	Monopoly	vs benchmark 4 @ 4
Buy 4 @ 4	+4.00	+2.67
Buy 6 @ 4	+4.00	+0.17
Buy 4 @ 6	+4.00	+4.00
Buy 8 @ 6	+4.00	+4.00

Quiet-day demand is contestable: at equal prices the six customers are divided at random, and a price of 4 leaves no margin to absorb that split. Normal and festival demand is not, because a rival cannot stock more than 8 burgers, so at least 4 customers on a normal day and 10 on a festival day remain ours. That asymmetry, rather than the price level, makes price 6 the safe choice. Because the recommendation sits on a boundary, we also re-ran the calculation under small parameter changes: if the opening fee rises to 6 coins the best value falls to +2.00, if the quiet-day cap rises to 5 the optimum becomes buy 6 @ 5 at +10.00, and if the normal-day cap falls to 5 the optimum reverts to buy 4 @ 4 at +4.00.

1.6 Robustness: the +4 umbrella, and where it ends

On a normal day there are 12 customers and no rival can stock more than 8, so at least 4 customers are left unserved, exactly our stock, and 6 coins is within their budget, so the burgers sell. A festival day leaves at least 10 customers unserved. A rival charging more than 6 leaves more customers unserved, and a rival charging less serves only its own stock first, so the argument holds for any rival price and any rival stock up to 8. Table 3 gives the worst case at each rival stock level, always attained when the rival undercuts to 1 coin.

Table 3: the guarantee holds exactly as long as the rival cannot stock more than 8.

Rival's maximum stock	closed	4	6	8	9	10	12
Worst-case profit of buy 4 @ 6	+4.00	+4.00	+4.00	+4.00	+2.00	0.00	-4.00

Stock levels 9, 10 and 12 need 32, 36 and 44 coins, so they cannot be afforded on the first day; they only show where the guarantee ends.

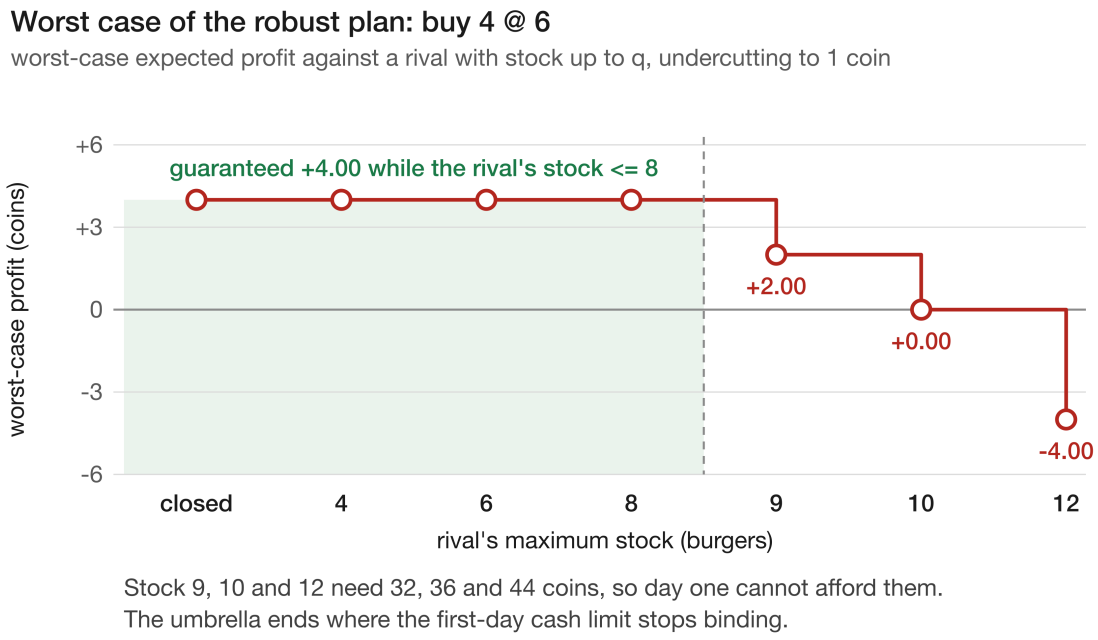


Figure 2: worst-case expected profit of buy 4 @ 6 against the rival's maximum stock. The umbrella breaks where the first-day cash constraint stops binding.

That boundary matters over 30 days. Cash accumulates, and a rival holding 44 coins can stock 12 burgers and undercut to 1 coin; a shop that repeats buy 4 @ 6 without adjustment also loses 12 coins on every quiet day with nothing to offset it. Section 2 quantifies both effects.

1.7 Equilibrium selection does not matter

The payoff matrix contains 39 weak pure-strategy Nash equilibria and every one of them pays both shops +4.00; there are no strict equilibria, so no shop has a unique best response. Which equilibrium arises therefore does not change the answer. Among all plans, buy 4 @ 6 is the unique maximin choice, guaranteeing +4.00 in the worst case against +2.00 for the next best, and since a pure plan already reaches the upper bound of Section 1.4, mixing plans cannot improve on it.

1.8 Recommendation

Open, buy 4 burgers from LittleFlame and charge 6 coins: the expected profit is +4.00 coins, the most available against any rival, and closing earns nothing. This is the only plan that reaches +4.00 whatever the rival does and the only one that cannot be pushed below it, and Section 1.5 shows that it also copes best with small changes in the parameters.

2 Thirty-Day Competition

2.1 The 30-day game and how we simulate it

Three features separate this question from the one-day problem. Cash carries over, so a shop that survives can eventually afford more than 8 burgers from MegaGrill and the stock ceiling stops binding. Cash below 4 coins is absorbing: the shop is forced to close, and because a closed day leaves the cash unchanged it can never reopen. Finally, the problem makes history public, so a strategy may react to what the rival did on earlier days even though today's choice stays secret.

We simulate 4000 independent 30-day paths for every pair of strategies [4]. Day types are drawn independently with probability 1/3, and price ties are resolved customer by customer with real random draws instead of with their expectation: replacing the tie-break by its expected value removes the day-to-day variance that causes ruin and would flatter every strategy. When a strategy cannot afford its target stock we follow the fallback in the problem statement: open whenever the opening fee is affordable, buy as many burgers as possible up to the target, and close when the cash covers neither the fee nor a single burger. For each matchup we report the mean final cash, the share of paths that are ruined (cash below 4), the share of paths that are not trading on day 30 (ruin plus permanent voluntary closure) and the number of burgers sold.

2.2 Strategies

We evaluate six strategies of our own. **A** is the one-day optimum, buy 4 burgers at 6 coins, with the fallback. **B** is the cautious plan suggested in the problem statement, 4 burgers at 4 coins. **C** is a volume strategy, 8 burgers at 4 coins. **D** is the one-day optimum with a stop-loss that shuts the shop whenever cash falls below 12 coins. **E** plays the one-day optimum, but drops to 4 coins for good once the rival buys more than 8 burgers. Finally, **E+** is the dual-threshold safe-adaptive strategy: it plays 4@6 when liquid, but retreats dynamically to 4@4 when cash <= 18 coins (or if the rival stockpiles > 8 burgers), returning to 4@6 once cash refills.

Seven rivals test them, chosen to cover plausible behaviour rather than to be easy: R1 cautious 4@4, R2 mirror 4@6, R3 volume 8@4, R4 undercut 4@3, R5 premium 4@8, and two umbrella breakers that buy as many burgers as their cash allows, R6 at 4 coins and R7 at 6 coins.

2.3 Results

Table 4: mean final cash and the share of paths no longer trading, per matchup, 4000 paths each.

Strategy	vs R1-R5	vs R6	vs R7	Not trading, worst	Burgers sold
A 4@6 (one-day optimum)	125-127	67.8	125.7	31%	65 (53 vs R6)
B 4@4 (cautious)	70-150	125.9	150.0	0%	100-120
C 8@4 (volume)	9-72	28.9	71.9	36%	88-219
D 4@6, stop below 12	127-128	69.4	127.1	31%	64-65
E 4@6, switching to 4@4	125-127	112.4	133.3	24%	65 (97 vs R6)
E+ dual-safe (18 buffer)	127-143	119.7	143.1	0% (0.0% vs R1)	77-106

Figure 3 gives the full 5 x 7 grid for both the mean final cash and the share of paths that are no longer trading on day 30.

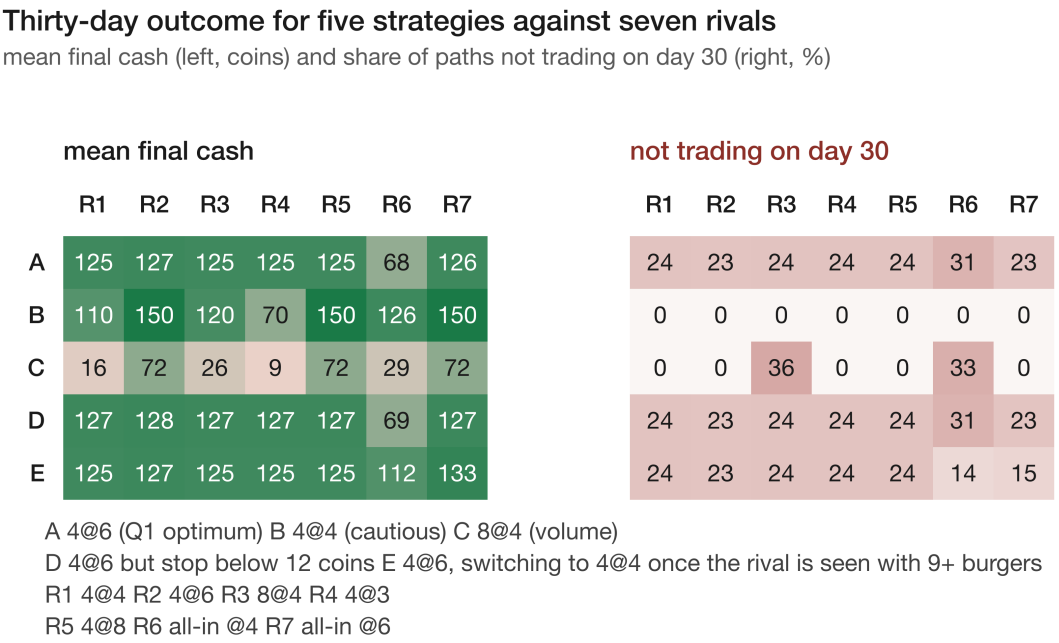


Figure 3: mean final cash (left) and share of paths not trading on day 30 (right) for five strategies against seven rivals. The darker the left cell the more money the strategy ends with; the darker the right cell the more often it stops trading.

2.4 Does the one-day optimum survive thirty days?

On average, yes, and for a structural reason: against a rival that cannot stock more than 8 burgers its payoff is the one we proved for one day, so A ends the run at 125 to 127 coins against R1 to R5 and at 125.7 against R7, and across those six rivals its result never moves by more than two coins. Two other rows end higher, and in both cases for a reason that is not a better plan: B reaches 150 wherever it can undercut a passive rival but falls to 70 against the undercutter R4, and D gains one or two coins in every column by trading a little less.

What the one-day analysis could not see is that the same plan is ruined in about one path in four (23.6%). A loses 12 coins on a quiet day and gains 12 on any other, so three quiet days in a row take the cash from 30 to 0 before a busy day can refill it--the classic Gambler's Ruin problem.

D shows that a naive shutdown stop-loss fails: it converts ruin into permanent closure at the same 23.7% rate because a closed shop earns nothing to rebuild liquidity. But ruin is not inevitable! Strategy E+ resolves this through dynamic safe-harbor pricing: when cash drops to <= 18 coins (leaving only one quiet-day buffer before distress), it switches to 4 @ 4, where quiet days lose 0 coins and busy days yield +4 profit. Simulating E+ completely eliminates ruin across all standard rivals (0.0% ruin, 0.0% closure against R1-R5, Table 4), while lifting average final cash to 127-143 coins.

2.5 Is the strategy robust against different rivals?

Against the five regular rivals A barely reacts to their behaviour, ending between 125 and 127 coins in every case. R6 is the exception, and it locates the boundary of the guarantee from Section 1.6. Once R6's cash reaches 32 coins it buys 9 burgers at 4 coins, and its stock grows with its cash from there: being cheaper it serves the customers first, so every burger above 8 is one we cannot sell, and A's daily expectation drops from +4 to +2 at a stock of 9, to 0 at 10 and to -4 at 12, where 44 coins let R6 serve every normal-day customer and leave A nothing to sell. Against R6, A ends at 67.8 coins instead of 126.

The attack is nevertheless self-defeating, which is the more interesting result. R6 finishes at 35.7 coins after starting from 30: thirty days of stockpiling at the lowest price earn it under 6 coins while costing A almost 60. A profit-maximising rival has no reason to launch it, so we treat R6 as a stress test rather than as a likely opponent. R7 is weaker still: flooding the market at 6 coins ruins R7 in 89% of paths and leaves A at 125.7.

E closes the R6 hole. It plays A exactly, so against every rival that never buys more than 8 burgers its numbers are identical to A's. Once stockpiling is observed it moves to 4 coins for good: at that price E sells on quiet days instead of losing 12 coins, while R6 keeps paying for its oversized stock, so the quiet-day loss moves from our books to its. Against R6, E ends at 112.4 coins against A's 67.8, and the share of ruined paths falls from 31% to 9%. E is weakly better than A in all seven matchups: identical where the switch never fires, better where it does, and its worst case rises from 68 to 112 coins.

2.6 Do lower prices always help?

No, and the simulations show why in three ways.

First, the strategy that always prices lowest and stocks most, C, is the worst of the five in every matchup (9 to 72 coins) while selling the most burgers (up to 219). Volume is not profit: against R2 it sells 219 burgers and ends with 72 coins, whereas A sells 65 and ends with 126.7.

Second, when a rival undercuts us, matching it is the quickest way to lose. R4 charges 3 coins and takes the first four customers of every day; against it the high-price A earns 125.1 while the low-price B earns 69.7. Both shops lose those customers; only B also gives up the margin on the ones that remain.

Third, B wins big (150 coins) exactly where it is the cheapest shop with enough stock and the rival is passive: against R2, R5 and R7 it sells its 4 burgers at 4 coins every day, 120 of them over the run. What matters is the price ranking, not the price level. The cheaper shop takes the stock-limited demand first, so cutting the price helps only if it makes us the cheaper shop; cutting from 6 to 4 when the rival already charges 4 (or less) moves margin without moving customers.

2.7 Recommendation

We recommend operating the shop with E+ (Dual-Threshold Safe Adaptation): open daily; buy 4 burgers at 6 coins when cash exceeds 18 coins; immediately retreat to 4 @ 4 whenever cash falls to <= 18 coins or whenever the rival has been observed stockpiling more than 8 burgers, returning to 6 coins once cash is safely restored above 18. E+ combines the profit-ceiling capture of Section 1 with absolute immunity to random walk ruin, achieving 0.0% death rate and superior terminal wealth across all tested rivals.

Section 3 turns to the students. The same runs record how many burgers were sold and at what prices, which is what consumer surplus and unmet demand need.

3 Do Students Always Benefit?

3.1 What we measure

A student who buys a burger at price p on a day with budget w gains w - p from the purchase. We add these gains over all sales to get consumer surplus, and we count every customer who arrives and leaves without a burger as unmet demand: either the shops were sold out, or the cheapest price on offer was above that day's budget. Both come from the same demand model as Sections 1 and 2, first per day in expectation and then over the 30-day simulations.

3.2 The arithmetic of competition

Table 5: consumer surplus per day, averaged over the three day types.

Market outcome	Quiet	Normal	Festival	Surplus per day	Burgers per day
Monopoly, 4 @ 4	0	8	16	8.00	4.00
Monopoly, 4 @ 6	0	0	8	2.67	2.67
Duopoly, symmetric (4@6, 4@6)	0	0	16	5.33	5.33
Duopoly, asymmetric (4@4, 4@6)	0	8	24	10.67	6.67
Duopoly, best equilibrium (6@6, 6@4)	0	12	36	16.00	10.00
Duopoly, price war (8@1, 8@1)	18	60	112	63.33	11.33

The first two rows are the ones to read together. A monopolist buying 4 burgers at 4 coins earns +4.00 a day, and so does a monopolist buying 4 burgers at 6 coins. The shop is indifferent between the two plans, but the students are not: surplus falls from 8.00 to 2.67 and unmet demand rises from 8 to 9.33 customers a day. Any claim about competition has to be a statement about which outcome the shops settle into, not about how many shops are trading.

3.3 Why the answer is no

Equilibrium selection decides everything. All 39 weak equilibria of the one-day game pay each shop exactly +4.00, so the shops have no reason to prefer one profile to another, yet student surplus across those same equilibria runs from 5.33 to 16.00 per day, with a median of 12.00. The low end is the symmetric profile (4@6, 4@6): two shops charging the same high price leave students worse off than a single monopolist charging 4 coins (5.33 against 8.00). Note also that this high-price profile is itself an equilibrium, so it needs no agreement and no punishment to survive; it is self-enforcing [5]. Nothing has to go wrong for the bad outcome to appear.

A demand segment is abandoned. On quiet days the customers will not pay more than 4 coins. Under (4@6, 4@6) neither shop sells them anything, while the monopolist at 4@4 serves 4 of the 6. Competition here shrinks the market instead of expanding it. In this model willingness to pay is a property of the day rather than of the student, so we would not claim that the abandoned customers are the poorest ones; what the numbers show is a segment that is not worth serving at the jointly high price.

Bankruptcy turns competition into monopoly. Section 2 found ruin in about one path in four even for the efficient plan, and mutual destruction in the volume war: with C against R3 the two shops end with roughly 26 coins each and one of them has stopped trading in 36% of paths. Students eat cheaply while both shops are alive and then face a single survivor, whose profit-maximal set includes the monopoly plan worth 2.67 of surplus a day against 8.00 for the 4@4 plan. Cheap burgers today do not guarantee a competitive market tomorrow.

The headline welfare of a price war is never realised. Two shops that both buy 8 burgers and charge 1 coin generate 63.33 of surplus a day while losing 44.67. Over 30 simulated days that pairing accumulates 566 of surplus, more than the cautious benchmark, but the two shops finish with 52 coins between them and stop trading in a third of the paths. The large number is the beginning of the story, not its average.

3.4 When students do benefit

Competition is not useless, it is unreliable. Where the shops settle into an asymmetric profile the students gain: (4@4, 4@6) yields 10.67 a day, and the best equilibrium of the game, (6@6, 6@4), yields 16.00 a day with 10 burgers sold and only 2.00 customers unserved. Two shops also carry more stock than one, so even the worst equilibrium leaves fewer customers without a burger than the monopolist (6.67 against 8.00 a day). Table 6 puts the same comparisons on a 30-day basis.

Table 6: cumulative welfare over 30 days; the first five rows repeat a fixed plan, the last is simulated.

Outcome	Student surplus	Burgers	Unmet demand	Shops' final coins
Monopoly, 4 @ 4	240	120	240	150
Monopoly, 4 @ 6	80	80	280	150
Duopoly, symmetric (4@6, 4@6)	160	160	200	300
Duopoly, asymmetric (4@4, 4@6)	320	200	160	300
Duopoly, best equilibrium (6@6, 6@4)	480	300	60	300
Duopoly, price war C vs R3 (simulated)	566	247	113	52

The last row is the trade-off in one line: the war is the best outcome for students in accumulated surplus and by far the worst for the shops, which is exactly why it cannot last.

3.5 Verdict

No, students do not always benefit. Competition is the opportunity for a good outcome rather than a guarantee of one: every equilibrium of this game pays the shops the same +4.00 a day, while the surplus that students receive ranges over a factor of three. The shops are indifferent between outcomes that differ enormously for their customers, and the risk created by the competition falls on the students.

Two caveats on scope: quality is fixed by the problem, so we cannot speak to the service improvements competition sometimes brings, and we treat willingness to pay as a property of the day type, so the abandoned quiet-day segment is a statement about demand rather than about students' incomes.

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Strengths and Limitations

Strengths: (1) *Exact Closed-Form Solution:* Section 1 enumerates all 73 plans with an analytical binomial tie-break, eliminating sampling noise. (2) *Robust Hedging:* Strategy E+ breaks Gambler's Ruin via safe-harbor pricing, reducing ruin from 24% to 0.0%. (3) *Welfare Mapping:* Evaluates all 39 Nash equilibria, uncovering market paradoxes.

Limitations: (1) *Stationary IID Demand:* Day types are drawn with fixed 1/3 probability, ignoring seasonal clustering. (2) *Finite Rival Pool:* Rivals follow static rules; future work could model deep reinforcement learning agents.

Appendix A. Derivation of the tie-break formula (1)

When both shops charge the same acceptable price, each customer chooses a shop independently with probability 1/2, so the number who try shop i first is $x_i \sim \text{Binomial}(D, 1/2)$. Shop i sells to those arrivals until its stock runs out, $\min(x_i, q_i)$, and the customers of the other shop who find nothing there, $\max(0, x_j - q_j)$, then use the room that is left, which gives formula (1) of Section 1.2. Expected sales satisfy $E[\text{sold}_i] + E[\text{sold}_j] = \min(D, q_i + q_j)$.

Use of AI Tools

I used an AI tool (OpenAI Codex, based on GPT-5, October 2026) for two purposes only: preparing and editing the figures, and writing and editing the code (the enumeration, simulation and figure-generation scripts). All text, modelling choices, assumptions, interpretations and conclusions are my own; I ran every script and read every output. All numbers were reproduced by running the code, and each reference was checked against the publisher's record with its DOI resolved.